



# Seminar

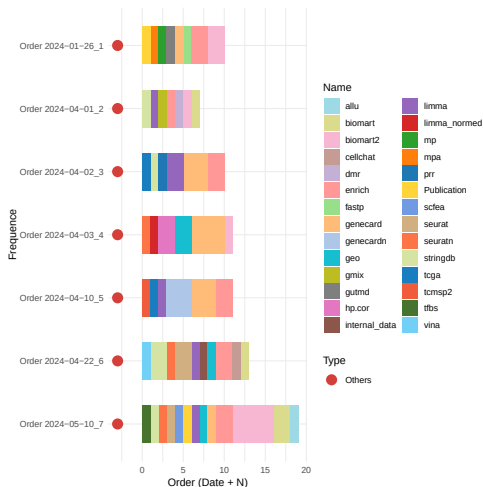
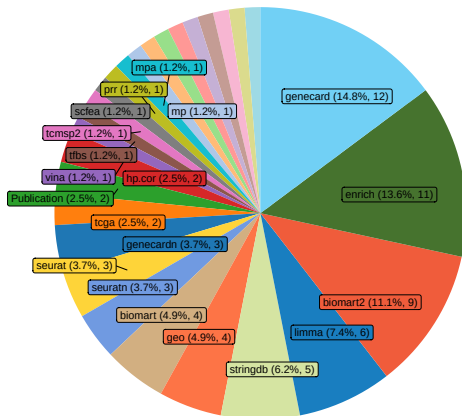
Lichuang Huang

Wie-Biotech

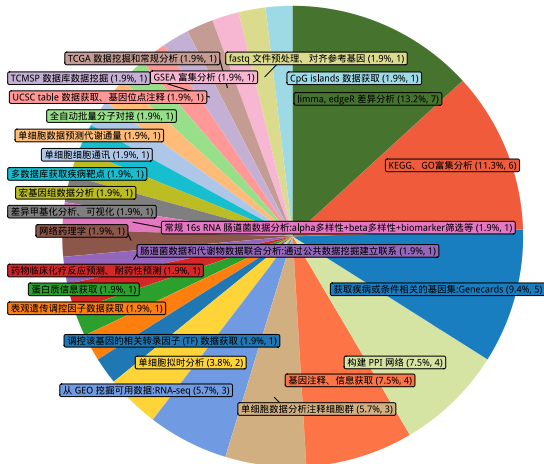
2024-06-14

## Tasks summary

# Overview 2024-05



# Methods 2024-05



# New Methods

- scfea: scFEA 单细胞数据的代写通量预测
- methy: missMethyl, DMRcate, Gviz Methyl-seq 数据分析
- ucscTable: rtracklayer UCSC Table 数据获取
- cig: makeCGI CpG Islands 预测
- dmr: Gviz DMR 数据可视化

# New Features report

- Auto Tags: 自动统计分析标签  $\Rightarrow$  Note
- Auto Section name: 自动生成分析标题, Method + Content + (Signature)

## Cardinal 空间代谢组数据分析 (INTEGRATE)

### ClusterProfiler 富集分析 (STNCANCER)

## Seurat 空间转录组分析 (CANCER)

- Delimiter



## Article sharing

# Theme

## A cross-package Bioconductor workflow for analysing methylation array data

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### Abstract

Methylation in the human genome is known to be associated with development and disease. The Illumina Infinium methylation arrays are by far the most common way to interrogate methylation across the human genome. This paper provides a Bioconductor workflow using multiple packages for the analysis of methylation array data. Specifically, we demonstrate the steps involved in a typical differential methylation analysis pipeline including: quality control, filtering, normalization, data exploration and statistical testing for probe-wise differential methylation. We further outline other analyses such as differential methylation of regions, differential variability analysis, estimating cell type composition and gene ontology testing. Finally, we provide some examples of how to visualise methylation array data.

# Visualization

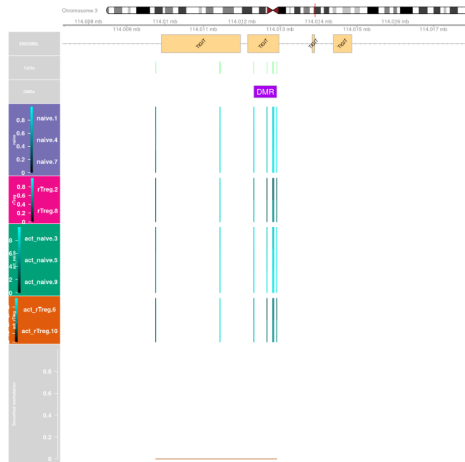


Figure 10: DMRcate allows you to quickly visualise DMRs in their genomic context

# Visualization

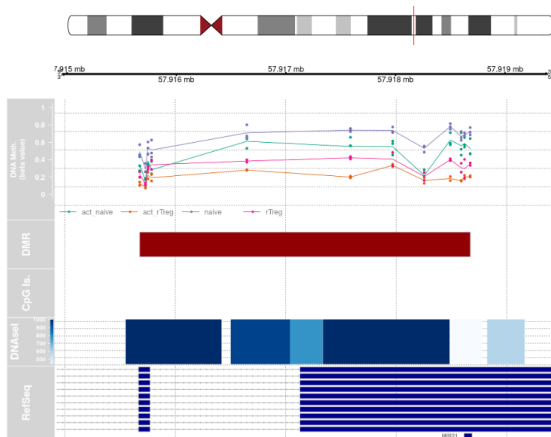


Figure 11: The Gviz package provides extensive functionality for customising plots of genomic regions. This plot shows one of the DMRs identified by the DMRcate analysis.

# Gviz for DNA

|   |                                              |
|---|----------------------------------------------|
| 1 | Introduction                                 |
| 2 | Basic Features                               |
| 3 | Plotting parameters                          |
| 4 | Track classes                                |
| 5 | Creating tracks from<br>UCSC data            |
| 6 | Track highlighting and<br>overlays           |
| 7 | Composite plots for<br>multiple chromosomes  |
| 8 | Bioconductor integration<br>and file support |
| 9 | SessionInfo                                  |
|   | References                                   |

## The Gviz User Guide

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**1 May 2024**

### Package

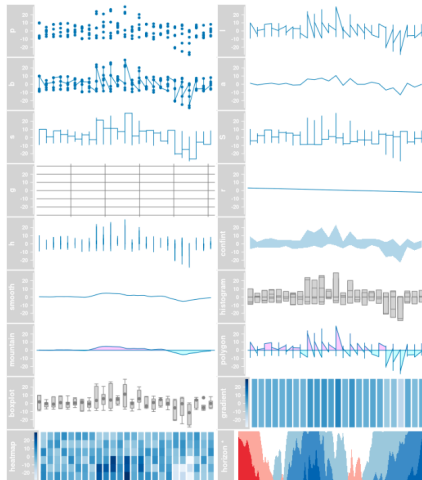
Gviz 1.48.0

**Last edited:** 19 May 2020

## 1 Introduction

# Gviz Visualization

- For Genome Visualization



# Schedule

# Schedule

- Finish orders
- New methods

Thanks